

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 08

DATE : 08-09-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 08

Topics

DATE : 08 - 09 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - II – PATTERN : [NEET]

PHYSICS : Vectors, Unit & Dimensions, Electrostatics, MEC, EMI (Till Induced Electric Field)

CHEMISTRY : Organic Chemistry (Till Taught)

BIOLOGY : Mendelian Genetics (Upto Sex Determination), Molecular Genetics (Upto Translation)
Human Health And Disease (Upto Immunology)

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

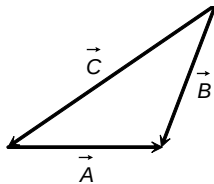
- Consider a co-axial cable which consists of an inner wire of radius a surrounded by an outer shell of inner and outer radii b and c respectively. The inner wire carries a current I and outer shell carries an equal and opposite current. The magnetic field at a distance x from the axis where $a < x < b$ is :-
 - (1) $\frac{\mu_0 I(c^2 - b^2)}{2\pi x(c^2 - a^2)}$
 - (2) $\frac{\mu_0 I(c^2 - a^2)}{2\pi x(c^2 - a^2)}$
 - (3) $\frac{\mu_0 I(c^2 - x^2)}{2\pi x(c^2 - b^2)}$
 - (4) Zero
- The magnetic field $d\vec{B}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i is :-
 - (1) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$
 - (2) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$
 - (3) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$
 - (4) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$
- A straight section PQ of a circuit lies along the x-axis from $x = -\frac{a}{2}$ to $x = \frac{a}{2}$ and carries a steady current i . The magnetic field due to the section PQ at a point $x = +2a$ will be :-
 - (1) Proportional to a
 - (2) Proportional to a^2
 - (3) Proportional to $1/a$
 - (4) Zero
- The magnetic field at the centre of an equilateral triangular loop of side $2L$ and carrying a current i is :-
 - (1) $\frac{9\mu_0 i}{4\pi L}$
 - (2) $\frac{3\sqrt{3}\mu_0 i}{4\pi L}$
 - (3) $\frac{2\sqrt{3}\mu_0 i}{\pi L}$
 - (4) $\frac{3\mu_0 i}{4\pi L}$
- The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ is
 - (1) 60°
 - (2) Zero
 - (3) 90°
 - (4) None of these
- If $|\vec{A} - \vec{B}| = |\vec{A}| = |\vec{B}|$, the angle between $\vec{A}$ and $\vec{B}$ is
 - (1) 60°
 - (2) 0°
 - (3) 120°
 - (4) 90°

SPACE FOR ROUGH WORK

7. Forces F_1 and F_2 act on a point mass in two mutually perpendicular directions. The resultant force on the point mass will be

- (1) $F_1 + F_2$ (2) $F_1 - F_2$
(3) $\sqrt{F_1^2 + F_2^2}$ (4) $F_1^2 + F_2^2$

8. For the figure



- (1) $\vec{A} + \vec{B} = \vec{C}$ (2) $\vec{B} + \vec{C} = \vec{A}$
(3) $\vec{C} + \vec{A} = \vec{B}$ (4) $\vec{A} + \vec{B} + \vec{C} = 0$

9. The value of the sum of two vectors $\vec{A}$ and $\vec{B}$ with θ as the angle between them is

- (1) $\sqrt{A^2 + B^2 + 2AB\cos\theta}$
(2) $\sqrt{A^2 - B^2 + 2AB\cos\theta}$
(3) $\sqrt{A^2 + B^2 - 2AB\sin\theta}$
(4) $\sqrt{A^2 + B^2 + 2AB\sin\theta}$

10. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$. Then the value of α is

- (1) -1 (2) $\frac{1}{2}$
(3) $-\frac{1}{2}$ (4) 1

11. Equal currents are flowing in three infinitely long wires along positive x, y and z directions. The magnetic field at a point $(0, 0, -a)$ would be (i = current in each wire) :-

- (1) $\frac{\mu_0 i}{2\pi a}(\hat{j} - \hat{i})$
(2) $\frac{\mu_0 i}{2\pi a}(\hat{i} - \hat{j})$
(3) $\frac{\mu_0 i}{2\pi a}(\hat{i} + \hat{j})$
(4) $\frac{\mu_0 i}{2\pi a}(-\hat{i} - \hat{j})$

12. A long straight wire carrying a current of 30A is placed in an external uniform magnetic field of induction 4×10^{-4} T. The magnetic field is acting parallel to the direction of current. The magnitude of the resultant magnetic induction in tesla at a point 2.0 cm away from the wire is ($\mu_0 = 4\pi \times 10^{-7}$ H/m):-

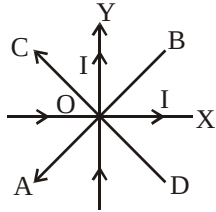
- (1) 10^{-4}
(2) 3×10^{-4}
(3) 5×10^{-4}
(4) 6×10^{-4}

13. If $\vec{A} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4\hat{k}$ then value of $|\vec{A} \times \vec{B}|$ will be

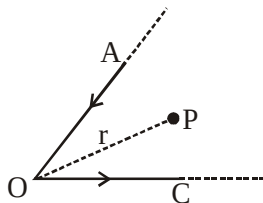
- (1) $8\sqrt{2}$ (2) $8\sqrt{3}$
(3) $8\sqrt{5}$ (4) $5\sqrt{8}$

SPACE FOR ROUGH WORK

14. Two very thin metallic wires placed along X and Y-axis carry equal currents as shown in figure. AB and CD are lines at 45° with the axes with origin of axes of O. The magnetic fields will be zero on the line :-



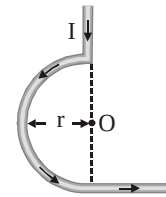
- (1) AB
(2) CD
(3) Segment OB only of line AB
(4) Segment OC only of line CD
15. Two parallel long wires carry currents i_1 and i_2 with $i_1 > i_2$. When the currents are in the same direction, the magnetic field midway between the wires is $10\mu\text{T}$. When the direction of i_2 is reversed, it becomes $40\mu\text{T}$. the ratio i_1/i_2 is :-
- (1) 3 : 4 (2) 11 : 7
(3) 7 : 11 (4) 5 : 3
16. Two wires AO and OC carry equal currents i as shown in figure. One end of both the wire extends to infinity. Angle AOC is α . The magnitude of magnetic field at a point P on the bisector of these two wires at a distance r from point O is :-



(1) $\frac{\mu_0 i}{2\pi r} \cot\left(\frac{\alpha}{4}\right)$ (2) $\frac{\mu_0 i}{4\pi r} \cot\left(\frac{\alpha}{2}\right)$

(3) $\frac{\mu_0 i}{2\pi r} \frac{\left(1 - \cos\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)}$ (4) $\frac{\mu_0 i}{4\pi r} \left(\frac{\alpha}{2}\right)$

17. In the figure, what is the magnetic field at the point O ?



(1) $\frac{\mu_0 I}{4\pi r}$ (2) $\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{2\pi r}$

(3) $\frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r}$ (4) $\frac{\mu_0 I}{4r} - \frac{\mu_0 I}{4\pi r}$

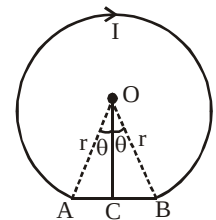
18. A wire is bent in the form of a circular arc with a straight portion AB. Magnetic induction at O when current I is flowing in the wire, is :-

(1) $\frac{\mu_0}{2r} (\pi - \theta + \tan \theta)$

(2) $\frac{\mu_0 I}{2\pi r} (\pi + \theta - \tan \theta)$

(3) $\frac{\mu_0 I}{2\pi r} (\pi - \theta + \tan \theta)$

(4) $\frac{\mu_0 I}{2\pi r} (-\tan \theta + \pi - \theta)$

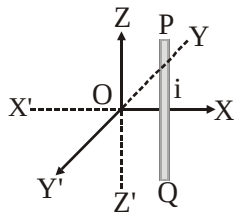


SPACE FOR ROUGH WORK

19. A proton (mass m and charge $+e$) and an α -particle (mass $4m$ and charge $+2e$) are projected with the same kinetic energy at right angles to the uniform magnetic field. Which one of the following statements will be true ?

- (1) The α -particle will be bent in a circular path with a small radius that for the proton
- (2) The radius of the path of the α -particle will be greater than that of the proton
- (3) The α -particle and the proton will be bent in a circular path with the same radius
- (4) The α -particle and the proton will go through the field in a straight line

20. A vertical wire kept in Z-X plane carries a current from Q to P (see figure). The magnetic field due to current will have direction at the origin O along :-

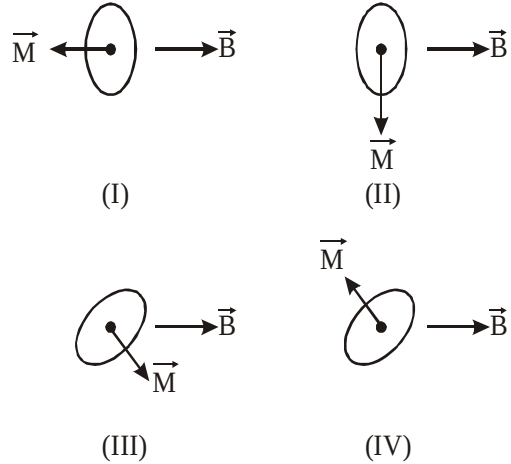


- (1) OX (2) OX' (3) OY (4) OY'

21. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω . The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on :-

- (1) ω and q (2) ω , q and m
- (3) q and m (4) ω and m

22. A current-carrying loop is placed in a uniform magnetic field in four different orientations. I, II, III and IV Fig. Arrange them in the decreasing order of potential energy :-



- (1) $I < III > II > IV$ (2) $I > II > III > IV$
- (3) $I > IV > II > III$ (4) $III > IV > I > II$

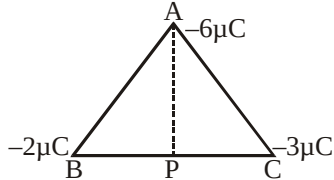
23. Two particles, each of mass m and charge q , are attached to the two ends of a massless rigid rod of length $2R$. The rod is rotated at constant angular speed about a perpendicular axis passing through its centre. The ratio of the magnitudes of the magnetic moment of the system and its angular momentum about the centre of the rod is :-

- (1) $\frac{q}{2m}$ (2) $\frac{q}{m}$ (3) $\frac{2q}{m}$ (4) $\frac{q}{\pi m}$

24. A magnet with moment P_m is given. If it is bent into a semicircular form, its new magnetic moment will be :-

- (1) P_m/π (2) $P_m/2$
- (3) P_m (4) $2P_m/\pi$

SPACE FOR ROUGH WORK

25. A current I flows along the length of an infinite long, straight thin walled pipe. Then :-
- The magnetic field at all points inside the pipe is the same, but not zero
 - The magnetic field is zero only on the axis of the pipe
 - The magnetic field is different at different points inside the pipe
 - The magnetic field at any point inside the pipe is zero
26. An electron is projected at an angle θ with a uniform magnetic field. If the pitch of the helical path is equal to the its radius, then the angle of projection is
- $\tan^{-1}\pi$
 - $\tan^{-1}2\pi$
 - $\cot^{-1}\pi$
 - $\cot^{-1}2\pi$
27. A conducting circular loop is placed in a uniform magnetic field of inducting B tesla with its plane normal to the field. Now, radius of the loop starts shrinking at the rate (dr/dt) . Then the induced e.m.f. at the instant when the radius is r is :-
- $\pi r B \left(\frac{dr}{dt} \right)$
 - $2\pi r B \left(\frac{dr}{dt} \right)$
 - $\pi r^2 B \left(\frac{dr}{dt} \right)$
 - $\frac{\pi B r^2}{2} \left(\frac{dr}{dt} \right)$
28. Mark the correct statement about the electric lines of force :-
- The number of lines of force leaving a +ve charge or entering a -ve charge is proportional to the magnitude of charge
 - The path of a charged particle in an electric field is not always along the field lines
 - The direction of electric lines of force is along the normal to the conducting surfaces
 - All of the above
29. If the electric field is given by $(5\hat{i} + 4\hat{j} + 9\hat{k})$, the electric flux through a surface of area 20 unit lying in the Y-Z plane will be :-
- 100 unit
 - 80 unit
 - 180 unit
 - 20 unit
30. ABC is equilateral triangle of side 1m. Charges are placed at its corners as shown in fig. P is the mid-point of side BC, the potential at point (P) is-
- 
- $2.7 \times 10^3 \text{ V}$
 - $1.52 \times 10^5 \text{ V}$
 - $1.3 \times 10^3 \text{ V}$
 - $-1.52 \times 10^5 \text{ V}$
31. The electron potential (V) as a function of distance (x) [in meters] is given by $V = (5x^2 + 10x - 9)$ volt. The magnitude of electric field at $x = 1\text{m}$ would be-
- 20 Volt/m
 - 6 Volt/m
 - 11 Volt/m
 - 23 Volt/m

SPACE FOR ROUGH WORK

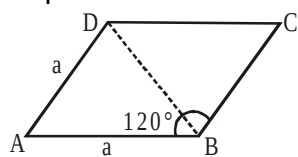
32. Which of the following is sufficient condition for finding the electric flux Φ_E through a closed surface?

- (1) If the magnitude of $\vec{E}$ is known everywhere on the surface
- (2) If the total charge inside the surface is specified
- (3) If the total charge outside the surface is specified
- (4) Only if the location of each point charge inside the surface is specified

33. Two infinitely long parallel wires having linear charge densities λ_1 and λ_2 respectively which are placed at a perpendicular distance of R metres. The force per unit length on either wire will be $\left(K = \frac{1}{4\pi\epsilon_0} \right)$:-

- (1) $K \frac{2\lambda_1\lambda_2}{R^2}$
- (2) $K \frac{2\lambda_1\lambda_2}{R}$
- (3) $K \frac{\lambda_1\lambda_2}{R^2}$
- (4) $K \frac{\lambda_1\lambda_2}{R}$

34. A charge $+Q$ at A (See figure) produces electric field E and electric potential V at D. If we now put charges $-2Q$ and $+Q$ at B and C respectively, then the electric field and potential at D will be-

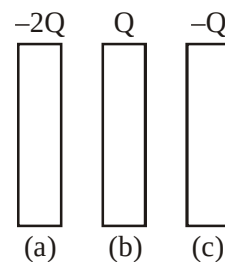


- (1) E and 0
- (2) 0 and V
- (3) $\sqrt{2}E$ and $\frac{V}{\sqrt{2}}$
- (4) $\frac{E}{\sqrt{2}}$ and $\frac{V}{\sqrt{2}}$

35. An electron (of charge $-e$) revolves around a long wire with uniform charge density λ in a circular path of radius r . Its kinetic energy is given by :-

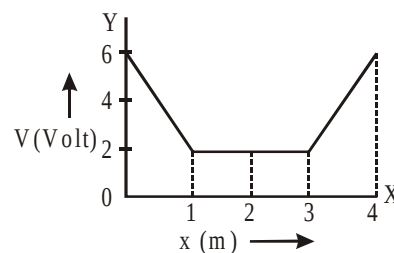
- (1) $\frac{\lambda e}{2\pi\epsilon_0 r}$
- (2) $\frac{\lambda e}{4\pi\epsilon_0 r}$
- (3) $\frac{\lambda e}{2\pi\epsilon_0}$
- (4) $\frac{\lambda e}{4\pi\epsilon_0}$

36. Three identical conducting plates are shown in figure. Find out the charge on the right face of the plate b.



- (1) $-Q$
- (2) $\frac{Q}{2}$
- (3) Q
- (4) Zero

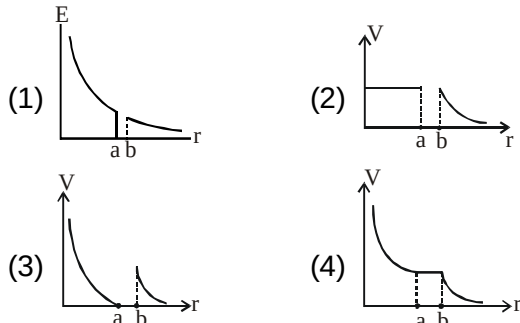
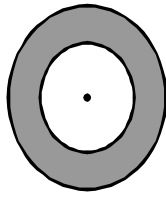
37. The variation of electric potential with distance from a fixed point is shown in figure. What is the value of electric field at $x = 0.5$ m (in N/C) is :



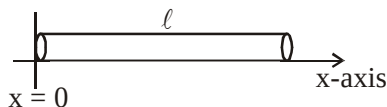
- (1) Zero
- (2) 3
- (3) 4
- (4) 2

SPACE FOR ROUGH WORK

38. Figure shows hollow metal sphere with internal radius a and external radius b . A point charge $+Q$ is present at the centre of sphere. Which of following graph represent the variation of potential with respect of distance from centre:-



39. If a proton, a deuteron and α -particle are kept in the same electric field :-
 (1) Proton and deuteron will have the same acceleration
 (2) Deuteron and α -particle will have the same acceleration
 (3) α -particle will have the maximum acceleration
 (4) Proton will have the minimum acceleration
40. Consider a rod of length ℓ having linear charge density ($\lambda = \lambda_0 x^2$) placed along x-axis whose one end is placed at origin. Find charge on the rod :-



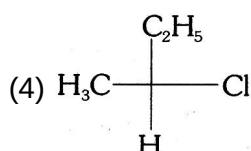
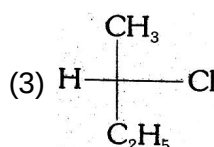
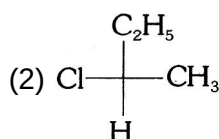
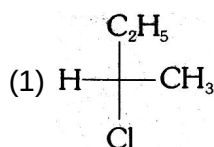
- (1) $\frac{\lambda_0 \ell^3}{3}$ (2) $\frac{\lambda_0 \ell^4}{4}$ (3) $\frac{\lambda_0 \ell^2}{2}$ (4) $\frac{\lambda_0 \ell^3}{4}$

41. If $x = at + bt^2$, where x is the distance travelled by the body in kilometre while t the time in seconds, then the units of b are
 (1) km/s (2) $km \cdot s$
 (3) km/s^2 (4) $km \cdot s^2$
42. A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b
 (1) MLT^{-3} and ML^2T^{-4}
 (2) MLT^{-3} and MLT^{-4}
 (3) MLT^{-1} and MLT^0
 (4) MLT^{-4} and MLT^1
43. The dimensions of K in the equation $W = \frac{1}{2} Kx^2$ is
 (1) $M^1L^0T^{-2}$ (2) $M^0L^1T^{-1}$
 (3) $M^1L^1T^{-2}$ (4) $M^1L^0T^{-1}$
44. Out of the following pair, which one does not have identical dimensions
 (1) Moment of inertia and moment of force
 (2) Work and torque
 (3) Angular momentum and Planck's constant
 (4) Impulse and momentum
45. If force (F), length (L) and time (T) are assumed to be fundamental units, then the dimensional formula of the mass will be

- (1) $FL^{-1}T^2$ (2) $FL^{-1}T^{-2}$
 (3) $FL^{-1}T^{-1}$ (4) FL^2T^2

SPACE FOR ROUGH WORK

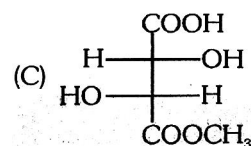
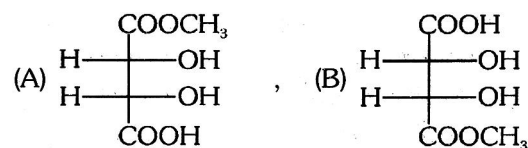
46. $\text{CH}_3-\text{CHCl}-\text{CH}_2-\text{CH}_3$ has a chiral centre which one of the following represents its R configuration



47. Which of the following compounds will exhibit cis-trans (geometrical) isomerism?

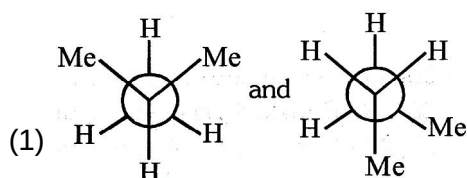
- (1) 1-Butanol (2) 2-Butene
(3) 2-Butanol (4) 2-Butyne

48. The correct statement about the compound (A), (B) and (C) is :-

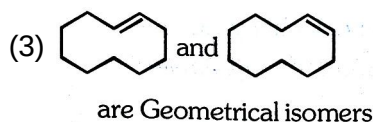
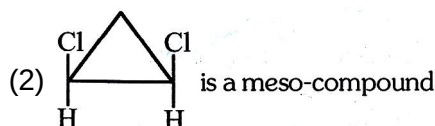


- (1) (A) and (B) are identical
(2) (A) and (B) are diastereomers
(3) (A) and (C) are enantiomers
(4) (A) and (B) are enantiomers

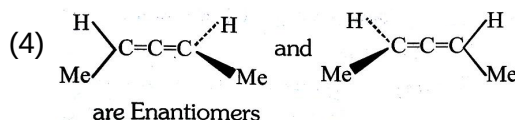
49. Which is incorrect statement:-



are conformations

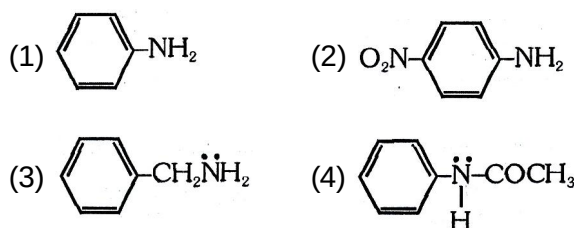


are Geometrical isomers

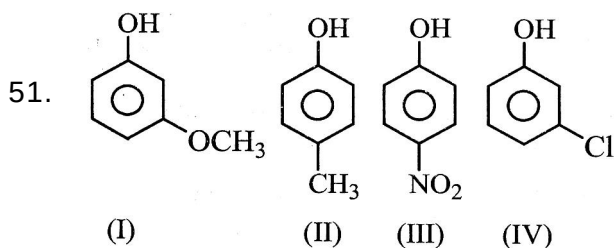


are Enantiomers

50. Which of the following compounds is most basic ?



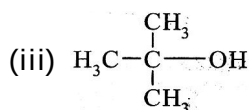
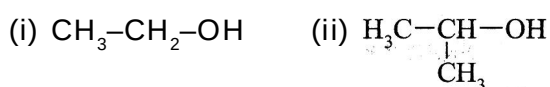
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The correct decreasing order of pK_a is:

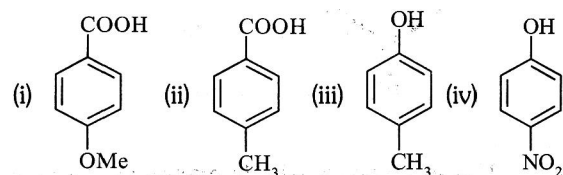
- (1) II > I > IV > III (2) IV > II > III > I
(3) III > II > IV > I (4) IV > I > II > III

52. Correct order of pK_a is



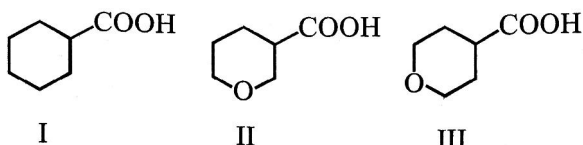
- (1) i > ii > iii (2) ii > i > iii
(3) iii > ii > i (4) ii > iii > i

53. Correct order of acidic strength is



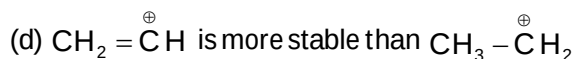
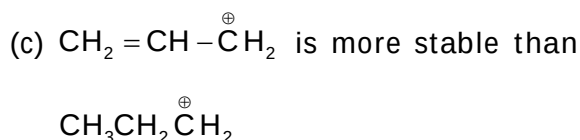
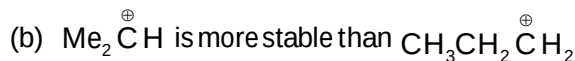
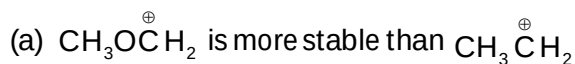
- (1) i > ii > iv > iii (2) iii > iv > ii > i
(3) ii > i > iv > iii (4) iv > ii > iii > i

54. The correct order of strengths of the carboxylic acids



- (1) III > II > I (2) II > I > III
(3) I > II > III (4) II > III > I

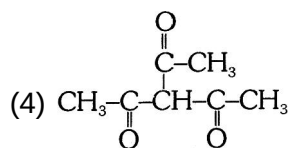
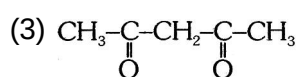
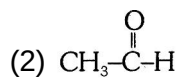
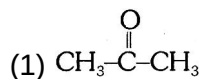
55. Consider the following statements :-



of these statements :-

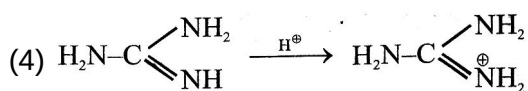
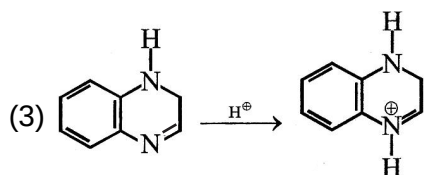
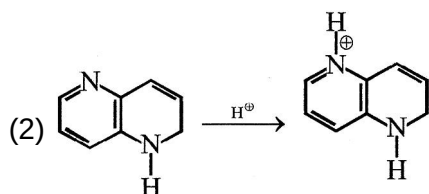
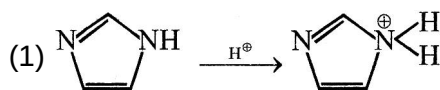
- (1) a and b are correct
(2) c and d are correct
(3) a, b and c are correct
(4) b, c and d are correct

56. Which of the following has the most acidic proton :-

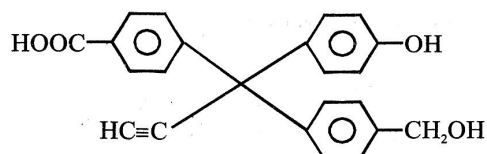


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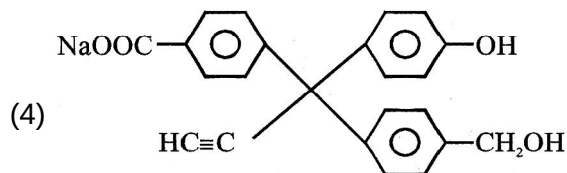
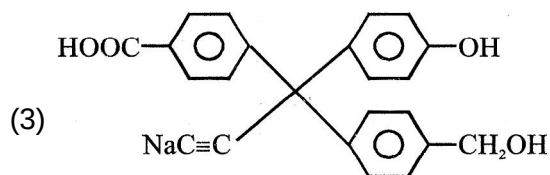
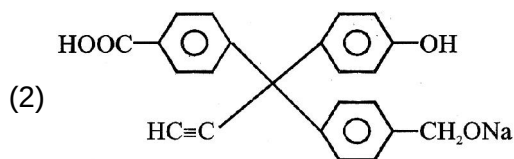
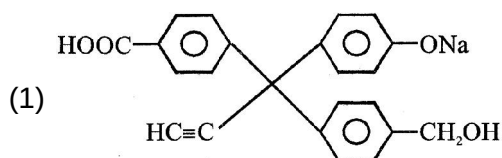
57. Which of the following acid-base reaction is not feasible :-



58.

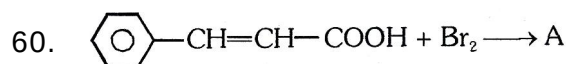


$\xrightarrow[\text{(1mole)}]{\text{NaNH}_2}$ product is :-



59. Which of the following orders of acid strength is correct :-

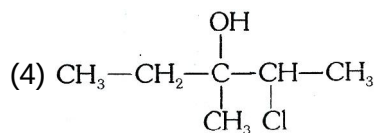
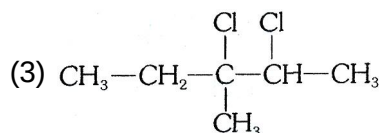
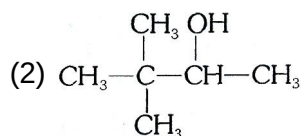
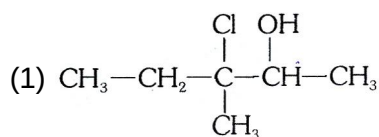
- (1) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC}\equiv\text{CH}$
- (2) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC}\equiv\text{CH}$
- (3) $\text{RCOOH} > \text{HOH} > \text{HC}\equiv\text{CH} > \text{ROH}$
- (4) $\text{RCOOH} > \text{HC}\equiv\text{CH} > \text{HOH} > \text{ROH}$



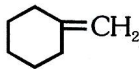
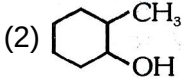
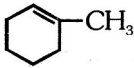
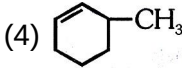
the number of chiral carbons in 'A' are

- (1) 1
- (2) 2
- (3) 3
- (4) 4

61. The predominant product formed when 3-methyl-2-pentene reacts with HOCl is

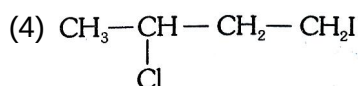
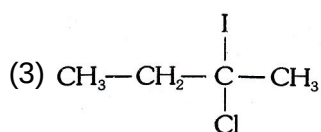
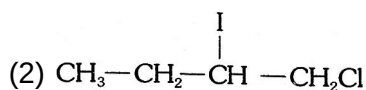
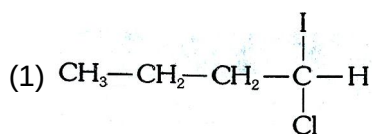
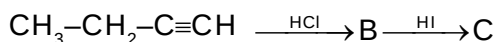


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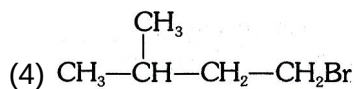
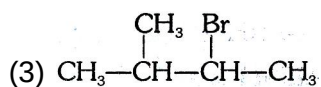
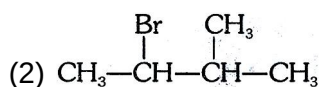
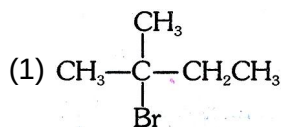
62. Cyanohydrin of the following compound on hydrolysis gives compound that can show optical isomerism :
- HCHO
 - CH₃CHO
 - CH₃COCH₃
 - All the above
63. Alkane reacts with which of the following halogens in dark
- F₂
 - Cl₂
 - I₂
 - Br₂
64. Only two isomeric monochloro derivatives are possible for (exclude stereo isomers)
- n-butane
 - 2, 4-dimethyl pentane
 - benzene
 - 2-methyl butane
65. The strongest deactivating effect on aromatic ring is
- CH₂Cl
 - OCH₃
 - CH₃
 - CCl₃
66. The function of anhydrous AlCl₃ in the Friedel craft's reaction
- To absorb water
 - To absorb HCl
 - To produce electrophile
 - To produce Nucleophile
67. The reaction,
CH₃Br + OH⁻ → CH₃OH + Br⁻ obeys the mechanism
- S_N1
 - S_N2
 - E₁
 - E₂
68. Most readily hydrolysed halide is
- C₆H₅Cl
 - (C₆H₅)₂CHCl
 - C₆H₅CH₂Cl
 - (C₆H₅)₃CCl
69. The least reactive chlorine is present in
- Methyl chloride
 - Allyl chloride
 - Ethyl chloride
 - Vinyl chloride
70. In the Williamson synthesis of ethers given by the general equation -
R—X + R'ONa → R—O—R' the yield from R—X follows the sequence :-
- CH₃X > 1° > 2° > 3°
 - CH₃X < 1° < 2° < 3°
 - CH₃X < 1° < 2° > 3°
 - CH₃X > 1° < 2° < 3°
71. The major product in the following reaction is
- $$\text{Cyclohexyl-CH}_2\text{-OH} \xrightarrow[\Delta]{\text{H}^+} ?$$
- 
 - 
 - 
 - 

SPACE FOR ROUGH WORK

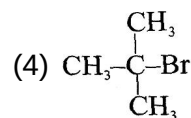
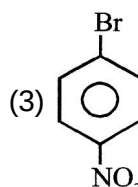
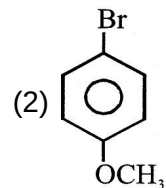
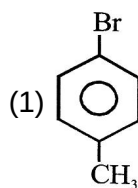
72. Predict the product 'C' obtained in the following reaction of 1-butyne :



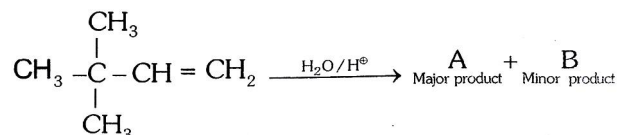
73. $\text{H}_3\text{C}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow \text{A}$
A (Predominantly) is :



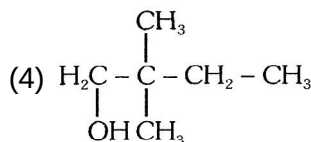
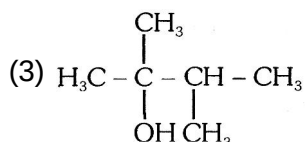
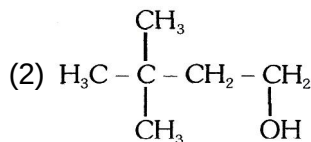
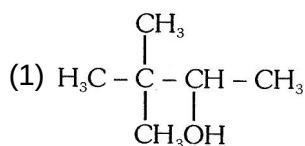
74. Which of the following is maximum reactive towards $\text{S}_\text{N}1$ reaction :-



75. In the following reaction:

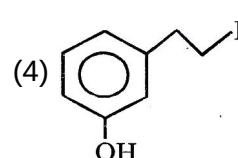
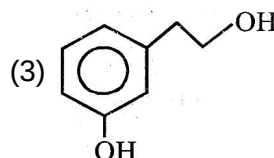
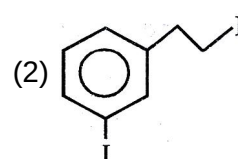
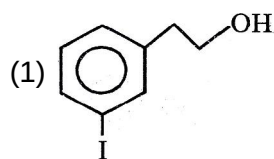
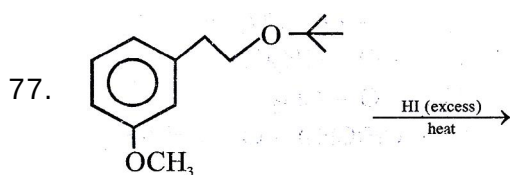
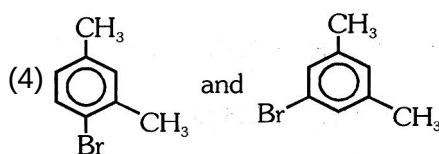
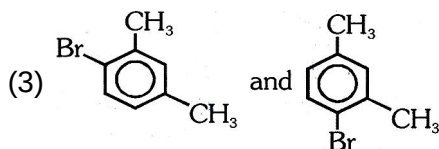
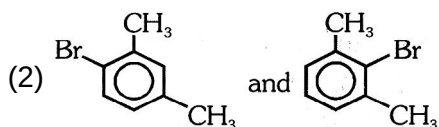
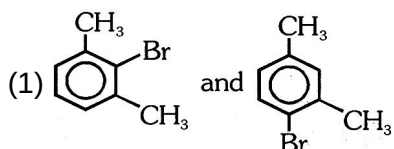
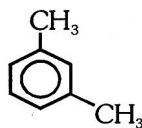


The major product is :-



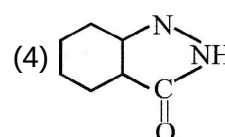
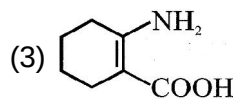
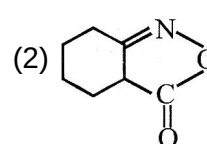
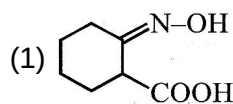
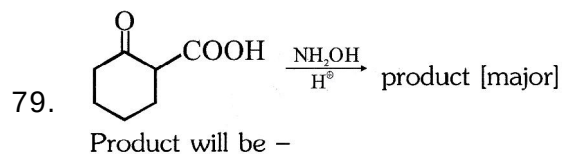
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76. What products are formed when the following compound is treated with Br_2 in the presence of FeBr_3 ?

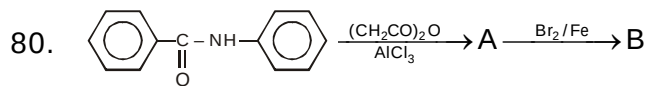


78. In an $\text{S}_{\text{N}}1$ reaction on chiral centres, there is :

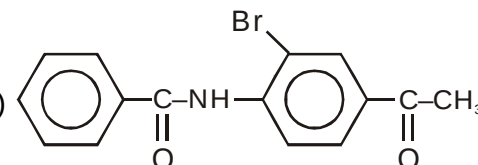
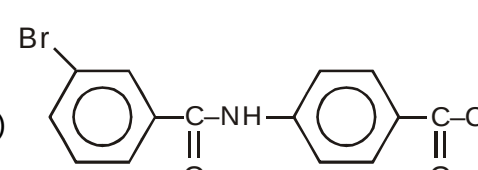
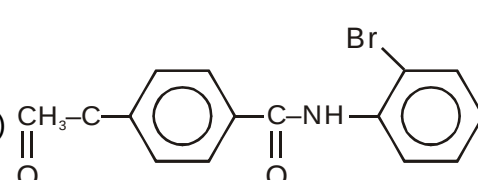
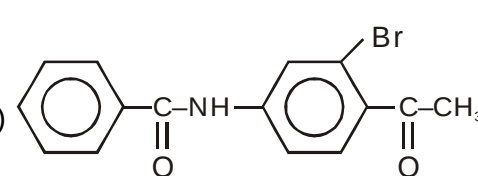
- (1) 100% retention
(2) 100% inversion
(3) 100% racemization
(4) inversion more than retention leading to partial racemization



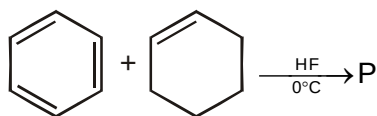
SPACE FOR ROUGH WORK



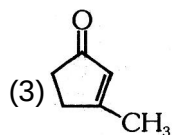
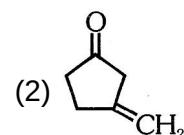
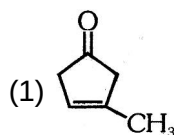
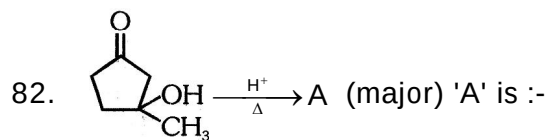
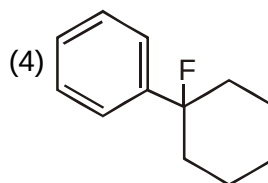
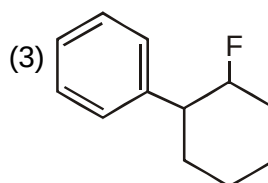
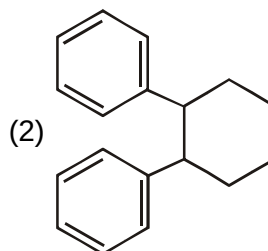
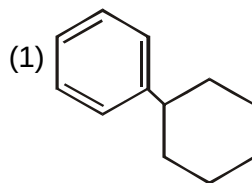
B is :-

- (1) 
- (2) 
- (3) 
- (4) 

81. In the given reaction



the product P is :-



(4) All of these

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83.  can be converted to  by

- (1) Red P + HI
- (2) Wolf Kishner reduction
- (3) Clemmensen reaction
- (4) All

84. The highest boiling point is expected for

- (1) Isooctane
- (2) 2,2,3,3-tetramethylbutane
- (3) n-octane
- (4) n-butane

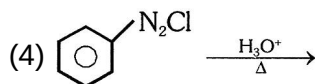
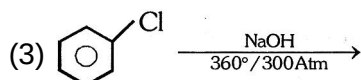
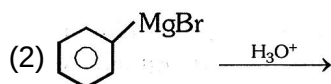
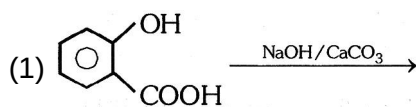
85. Ozonolysis of 3-Methyl-1-butene gives a mixture of

- (1) Propanal and ethanal
- (2) Propanone and ethanal
- (3) 2-Methylpropanal and methanal
- (4) Butanone and methanal

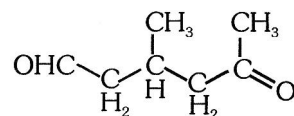
86. When treated with ammonical cuprous chloride, which one among the following forms red precipitate ?

- (1) C_2H_6
- (2) C_2H_4
- (3) C_2H_2
- (4) C_6H_6

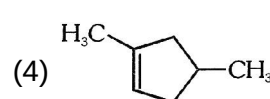
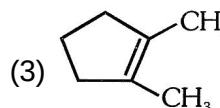
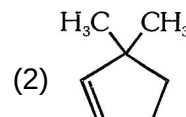
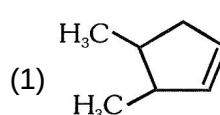
87. In which of the following reactions benzene is obtained



88. A single compound of the structure :-

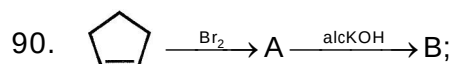


is obtainable from ozonolysis of which of the following cyclic compounds?

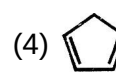
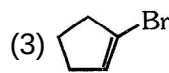
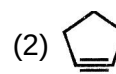
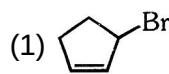


89. The volume of methane evolved by treatment of 16.6 g of methyl magnesium iodide with water at S.T.P. is

- (1) 224 mL
- (2) 2.24 L
- (3) 0.224 L
- (4) 22.4 L



What is the structure of B



SPACE FOR ROUGH WORK

91. During transcription, RNA polymerase binds to DNA sequence and DNA assumes a saddle like structure at that point. The sequence is called
- AAAT box
 - TATA box
 - GGTT box
 - CAAT box
92. RNA polymerase requires for initiation
- Sigma subunit
 - α -subunit
 - rho subunit
 - Spliceosome
93. Which one is correctly matched?
- AUG,ACG—Start or methionine
 - UUA, UCA—Leucine
 - GUU—Alanine
 - UAG, UGA—Stop
94. Removal of introns and joining the exons in a defined order in a transcription unit is
- Splicing
 - Capping
 - Tailing
 - Transformation
95. Whose experiments cracked DNA and discovered triplet nature of genetic code?
- Beadle and Tatum
 - Hershey and Chase
 - Morgan and Sturtevant
 - Nirenberg and Mathaei
96. Site of tRNA that binds to mRNA molecule is
- 3'end
 - 5'end
 - Codon
 - Anticodon
97. Which is not a salient feature of genetic code?
- Ambiguous
 - Universal
 - Specific
 - Degenerate
98. Which is not part of transcription unit?
- Promoter
 - Terminator
 - Structural gene
 - Inducer
99. Select the correct option.
- | Direction of RNA synthesis | Direction of reading of the template DNA strand |
|-----------------------------|---|
| (1) 3' $\longrightarrow$ 5' | 3' $\longrightarrow$ 5' |
| (2) 5' $\longrightarrow$ 3' | 3' $\longrightarrow$ 5' |
| (3) 3' $\longrightarrow$ 5' | 5' $\longrightarrow$ 3' |
| (4) 5' $\longrightarrow$ 3' | 5' $\longrightarrow$ 3' |
100. Heterogeneous nuclear RNA is converted into mRNA by
- Splicing
 - Capping
 - Tailing
 - All of the above
101. Which of the following nitrogenous base is double ringed?
- Guanine
 - Thymine
 - Uracil
 - Cytosine
102. Which one of the following is the starter codon?
- | | |
|---------|---------|
| (1) UAA | (2) UAG |
| (3) AUG | (4) UGA |
103. Wobble hypothesis establishes
- Peptide chain formation
 - Initiation of peptide chain
 - Termination of peptide chain
 - Economy in tRNA molecules

104. Find out the correct answers out of the following discoveries.

1. Griffith—Transformation
2. Gamow—Triplet code
3. Miescher—Nucleic acid

- (1) 1, 2 and 3 correct
- (2) 1 and 2 correct, 3 false
- (3) 1 correct, 2 and 3 false
- (4) 1 and 3 correct, 2 false

105. Protein helping in opening of DNA double helix in front of replication fork is

- (1) DNA gyrase
- (2) DNA polymerase-I
- (3) DNA ligase
- (4) DNA topoisomerase

106. tRNA recognises ribosome by

- (1) T Ψ C loop
- (2) DHU loop
- (3) Anticodon
- (4) AA-site

107. DNA produced from RNA is called

- (1) A-DNA
- (2) B-DNA
- (3) Z-DNA
- (4) c-DNA

108. Match the columns I and II, and choose the correct combination from the options given.

Column I		Column II	
(a)	F. Griffith	(1)	<i>E. coli</i>
(b)	Hershey and Chase	(2)	Bacteriophage
(c)	Meselson and Stahl	(3)	<i>Vicia faba</i>
(d)	Taylor stahl	(4)	<i>Pneumococcus</i>

Options:

- (1) a – 4, b – 1, c – 2, d – 3
- (2) a – 4, b – 2, c – 1, d – 3
- (3) a – 2, b – 1, c – 3, d – 4
- (4) a – 4, b – 3, c – 2, d – 1

109. Read the following statements and choose the correct option.

- (A) RNA polymerase associates transiently with 'Rho' factor to initiate tanscription.
- (B) In bacteria transcription and translation takes place in the same compartment.
- (C) RNA polymerase I is responsible for transcription of tRNA.
- (D) When hnRNA undergoes capping process, adenylate residues are added at 3' end in a template independent manner.
- (E) hnRNA is precursor of mRNA

Options:

- (1) A, B and E
- (2) B, C, D and E
- (3) B and E
- (4) A, B, C and D

110. Study the lists and find the correct match

List I		List II	
(a)	Exon	(I)	Site for binding of RNA polymerase
(b)	Capping	(II)	Coding sequence
(c)	Tailing	(III)	Lagging strand
(d)	Promoter	(IV)	Methyl guanosine triphosphate
		(V)	Adenylate residues

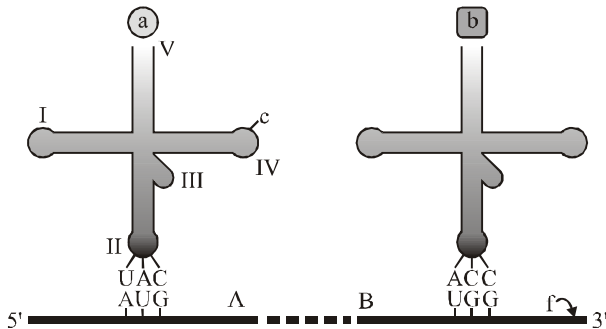
Options:

- (1) a – II, b – IV, c – V, d – I
- (2) a – II, b – IV, c – I, d – V
- (3) a – III, b – I, c – II, d – IV
- (4) a – IV, b – II, c – III, d – I

111. Which one is not transcribed by DNA?

- (1) Exons
- (2) Introns
- (3) DNA
- (4) tRNA

112. Recognise the figure and find out the correct matching.



	A	B
1	Met	Tryptophan
2	Val	Tyrosine
3	Glucine	Tryptophan
4	Met	Tyrosine

113. Codon with dual function is

- (1) UGA (2) UUU
(3) AUG (4) AAA

114. rRNA that catalyse the formation of peptide bond in eukaryotes is synthesised by

- (1) RNA Polymerase I
(2) RNA Polymerase II
(3) RNA Polymerase-III
(4) RNA -Polymerase

115. (A) The experiments which proved that DNA in chromosome also replicate semiconservatively was performed on ____i____.

(B) Genetic material should provide the scope for slow changes that are required for ____ii____

(C) In DNA plane of one base pair stacks over the other in double helix confer additional ____iii____ to the helicle structure.

(D) An acidic substansce present in nucleus was named ____iv____ by Friedrich meischer.

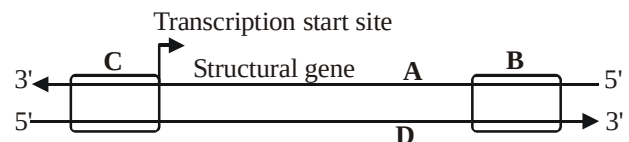
	I	II	III	IV
(1)	<i>Vicia faba</i>	Evolution	Stability	Nuclein
(2)	<i>Entamoeba coil</i>	Stability	Flexibility	Nucleic acid
(3)	<i>Entamoeba coil</i>	Evolution	Stability	Nuclein
(4)	<i>Vicia faba</i>	Mutation	Flexibility	Nucleic acid

116. Find out the correct match

a	RNA Polymerase-I	(i)	28 S rRNA
b	RNA Polymerase-II	(ii)	5.8 S rRNA
c	RNA Polymerase-III	(iii)	tRNA
		(iv)	hn RNA
		(v)	18 S rRNA
		(vi)	Sn RNA
		(vii)	5 S rRNA
		(viii)	mRNA

	a	b	c
(1)	I, II, V, VII	IV, VIII	III, VI
(2)	I, II, VII	IV, VI, VIII	III, V
(3)	I, II, V	IV, VIII	III, VI, VII
(4)	V, VII, II	I, IV, VIII	III, VI

117. In the given schematic structure of a transcription unit, identify the correct labelling for A to D.



	A	B	C	D
(1)	Promotor	Template strand	Terminator	Coding strand
(2)	Template strand	Terminator	Promotor	Coding strand
(3)	Coding strand	Terminator	Promotor	Template strand
(4)	Template strand	Promotor	Terminator	Coding strand

118. DNA replication is :-

- (1) Autocatalytic function of DNA
- (2) Semiconservative process
- (3) Semidiscontinuous process
- (4) All of these

119. In a newly synthesized DNA strand the required sequence of nucleotides is as follows :-

3' ATTCGAGTCCAGCTCA 5'

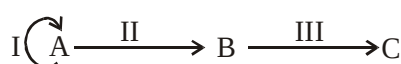
Then what will be the nucleotide sequence in template strand?

- (1) 3' ATTCGAGTCCAGCTCA 5'
- (2) 5' ACTCGACCTGAGCTTA 3'
- (3) 3' TGAGCTGGACTCGAAT 5'
- (4) 5' ATTCGAGTCCAGCTCA 3'

120. Which of the following is synthesized by RNA polymerase in prokaryotes ?

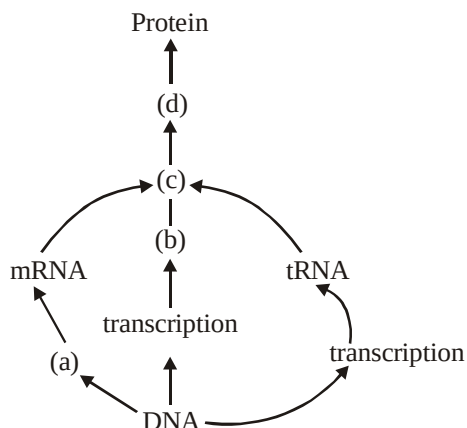
- (1) 28s RNA (2) 5S rRNA
- (3) 18s rRNA (4) 5.8S rRNA

121. Francis crick proposed the central dogma in molecular biology which state flow of genetic information as :-



- | | | |
|-----------------|------------------|-----------------|
| (1) A m RNA | B Protein | C DNA |
| I Transcription | II Replication | III Translation |
| (2) A DNA | B m RNA | C Protein |
| I Replication | II Transcription | III Translation |
| (3) A DNA | B Protein | C m RNA |
| I Replication | II Transcription | III Translation |
| (4) A m RNA | B DNA | C Protein |
| I Transcription | II Replication | III Translation |

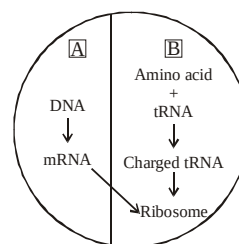
122. Identify (a), (b) (c) and (d)



Options :-

	(a)	(b)	(c)	(d)
(1)	transcription	rRNA	ribosome	translation
(2)	transcription	genetic code	ribosome	translation
(3)	replication	rRNA	ribosome	translation
(4)	transcription	rRNA	nucleus	translation

123. Represented below is the event in transfer of genetic information. Which one of the following option is correct according to the given pattern?



- (1) A-Cytoplasm, B-Nucleus
- (2) A-Nucleus, B-Cytoplasm
- (3) A-Mitochondria, B-Nucleus
- (4) A-Nucleus, B-Mitochondria

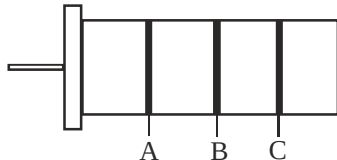
124. Identify the correct match from the column I, II and III.

	Column I		Column II		Column III
1.	DNA polymerase I	a.	Work on Bacteriophage	i	DNA Repair
2.	Meselson and Stahl	b.	Form mRNA	ii	Use N isotope
3.	RNA polymerase II	c.	Remove RNA primer	iii	Prove genetic material is DNA
4.	Hershey and Chase	d.	Proved semi Conservative method	iv.	DNA dependant RNA Polymerase

- (1) 1- b- iv, 2 -d - ii, 3-a-i, 4 -a - iii
- (2) 1- c- i, 2 -d - ii, 3-b-iv, 4 -a - iii
- (3) 1- c- i, 2 -a - iii, 3-b-iv, 4 -d - ii
- (4) 1- c- i, 2 -d - iii, 3-b-iv, 4 -a - ii

125. The discontinuously synthesised fragments are later joined by the enzyme
- (1) DNA P I
 - (2) DNA Ligase
 - (3) Primase
 - (4) Topoisomerase

126. Consider the diagram given below :-



The correct match is :-

- (1) A – $^{15}\text{N} - ^{15}\text{N}$, B – $^{15}\text{N} - ^{14}\text{N}$, C – $^{14}\text{N} - ^{14}\text{N}$
 - (2) A – $^{15}\text{N} - ^{14}\text{N}$, B – $^{15}\text{N} - ^{15}\text{N}$, C – $^{14}\text{N} - ^{14}\text{N}$
 - (3) A – $^{14}\text{N} - ^{14}\text{N}$, B – $^{15}\text{N} - ^{14}\text{N}$, C – $^{15}\text{N} - ^{15}\text{N}$
 - (4) A – $^{14}\text{N} - ^{14}\text{N}$, B – $^{15}\text{N} - ^{15}\text{N}$, C – $^{14}\text{N} - ^{15}\text{N}$
127. What % of trihybrid will be obtained by the cross of trihybrid plant with a recessive plant :-
- (1) 25%
 - (2) 12.5%
 - (3) $\frac{1}{64} \times 100$
 - (4) $\frac{3}{8} \times 100$
128. If father has pseudorickets and mother is normal then :-
- (1) All offspring will be affected
 - (2) Only male child will be affected
 - (3) Only female child will be affected
 - (4) One male and female child will be affected
129. If a sperm of grasshopper fertilize egg of grasshopper then the zygote can be :-
- (1) Female XX
 - (2) Male XY
 - (3) Male XX
 - (4) Female XO

130. Which is related to Barr body :-

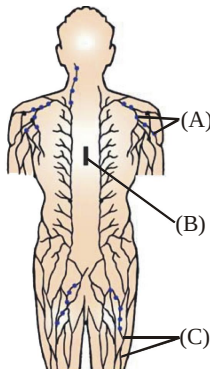
- (1) Y-body
 - (2) Lyon hypothesis
 - (3) Schwann hypothesis
 - (4) Genic-balance theory
131. Morgan and his group found that when genes were grouped on the same chromosome, some genes were very tightly linked and showed :-
- (1) Very low recombination
 - (2) Higher recombination
 - (3) No recombination
 - (4) 100% parental combination
132. The recombination frequency between gene A and B is 15%. What is the distance between gene A and B ?
- (1) 5 cm
 - (2) 15 cm
 - (3) 10 cm
 - (4) Cannot determined
133. A holandric gene causes hypertrichosis. When a man with hairy pinna marries a normal women, what percentage of their daughter would be expected to have hairy ears ?
- (1) 50%
 - (2) 25%
 - (3) 75%
 - (4) 0%
134. In a cross performed by Morgan in *Drosophila*, the number of offsprings obtained were as follows :
- $$W^+m^+ = 31.4\%, Wm = 31.4\%$$
- $$+ m = 18.6\%, w + = 18.6\%$$
- Find out the distance between gene W and m.
- (1) 37.2
 - (2) 18.6
 - (3) 62.8
 - (4) 31.4
135. Complete linkage is found in :
- (1) Male *Drosophila*
 - (2) Female *Silkworm*
 - (3) *Lathyrus odoratus*
 - (4) Both (1) and (2)

136. Antigen binding site in an antibody is found between
 (1) Two light chains
 (2) Two heavy chains
 (3) One heavy and one light chain
 (4) Both 2 and 3 depending upon nature of antigen.

137. In higher vertebrates, the immune system can distinguish self-cells and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to

- (1) Auto-immune disease
 (2) Active immunity
 (3) Allergic response
 (4) Graft rejection

138. In given diagramme identify (A), (B) and (C)



- (1) (A) – Blood vessels, (B) – Tonsils, (C) – Lymph nodes
 (2) (A) – Lymphatic vessels, (B) – Bone marrow, (C) – Thymus
 (3) (A) – Lymph nodes, (B) – Thymus, (C) – Lymphatic vessels
 (4) (A) – Payer's patches, (B) – Spleen, (C) – Blood vessels

139. Bursa Fabricius of birds is associated with
 (1) Generation of basophils
 (2) Generation of B-cells
 (3) Metabolism of fatty acids
 (4) Production of uric acid

140. A person is injected with gammaglobulin against hepatitis B. it is
 (1) Artificially acquired active immunity
 (2) Naturally acquired active immunity
 (3) Artificially acquired passive immunity
 (4) Naturally acquired passive immunity

141. Cells involved in specific immune mechanism are?

- (1) Erythrocyte (2) Lymphocyte
 (3) Eosinophills (4) Thrombocytes

142. Migration of phagocytes and other leucocytes towards the place of infection occurs through

- (1) Thrombosis (2) Diapedesis
 (3) Emboly (4) Phagocytosis

143. Mark out the correct statement regarding the natural killer cells–

- (a) They are a part of innate cellular barrier.
 (b) NK cells release perforin during their interaction with the target cells.
 (c) NK cells destroy both tumours and virally infected cells.
 (d) NK cells are a type of lymphocytes.

- (1) a, b, c, d (2) b, c, d only
 (3) a only (4) a, b only

144. Diseases which are easily transmitted from one person to another, are called

- (1) Infectious diseases
 (2) Non-infectious diseases
 (3) Deficiency diseases
 (4) Genetic disorders

145. In immunoglobulin disulphides do not join

- (1) Two heavy chains
 (2) Light chain with heavy chain
 (3) Two light chains
 (4) All of the above

146. Each immunoglobulin has two heavy chains and two light chains. The antigen binding site is found in

- (1) Variable region of heavy chain
 (2) Variable region of light chain
 (3) Constant region of light chain
 (4) Variable region of both heavy and light chain

147. Damage to thymus in a child would lead to

- (1) Loss of cell mediated immunity
 (2) Loss of antibody mediated immunity
 (3) A reduction in stem cell production
 (4) Reduction in haemoglobin content of blood

148. Active natural immunity means

 - (1) Resistance developed before disease
 - (2) Resistance developed after disease
 - (3) Increased heart beat
 - (4) Increased flow of blood

149. Antigens are found

 - (1) Inside cytoplasm
 - (2) Inside nucleus
 - (3) On nuclear envelope
 - (4) On cell surface

150. Memory-based acquired immunity evolved in

 - (1) Higher vertebrates
 - (2) Lower vertebrates
 - (3) Lower invertebrates
 - (4) All of the above

151. Incorrect about spleen is -

 - (1) Secondary lymphoid organ
 - (2) Bean - shaped
 - (3) Contains phagocytes only
 - (4) Traps blood - borne micro organisms

152. Which cells has phagocytic nature ?

 - (A) PMNL - neutrophil
 - (B) Monocyte
 - (C) NK cells
 - (1) A, B (2) B
 - (3) A, B, C (4) B, C

153. Fill in the blanks:

 1. The ...a... lymphoid organs provide the sites for interaction of lymphocytes with antigen, which then proliferate to become ...b...
 2. The thymus is a lobed organ located near the heart and beneath the ...C...
 3. Spleen has a large reservoir of ...d...
 - (1) a—primary, b—antigen sensitive cells, c—collar bone, d—lymphocytes
 - (2) a—secondary, b—receptor cells, c—breastbone, d—erythrocytes
 - (3) a—primary, b—effector cells, c—breast bone, d—lymphocytes
 - (4) a—secondary, b—effector cells, c—breastbone, d—erythrocytes

154. Which portion of an antibody provides antigen-binding sites?

 - (1) Light chain
 - (2) Constant portion
 - (3) Heavy chain
 - (4) Variable portion

155. The overall ability of the to fight the by a system is called :

 - (1) host, pathogen, immunity
 - (2) pathogen, host, immunisation
 - (3) Immunity, host, pathogen
 - (4) host, immunity, vaccination

156. Innate immunity is :

 - (1) Specific
 - (2) Non-specific
 - (3) Sometimes specific
 - (4) (2) & (3) both

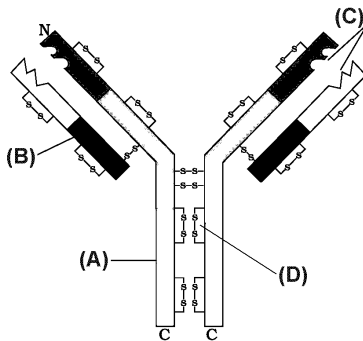
157. Which of the following is not one of the three main antigen-presenting cell types?

 - (1) Dendritic cell
 - (2) B-Lymphocytes
 - (3) Natural killer cell
 - (4) Macrophages

158. PMNL and natural killer cells are example of which type of barrier :

 - (1) Physiological
 - (2) Cellular
 - (3) Physical
 - (4) None

159. Identify the structure A, B, C & D in antibody molecule:



- (1) A = Heavy Chain, B = Light chain, C = Antigen binding site, D = disulphide bond
- (2) A = Heavy Chain, B = Light chain, C = disulphide bond, D = Antigen binding site
- (3) A = Light chain, B = Heavy Chain, C = disulphide bond, D = Antigen binding site
- (4) A = Antigen binding site, B = Heavy Chain, C = disulphide bond, D = Light chain

160. Which immune response is quick and highly intensified:

- (1) Primary
- (2) Secondary
- (3) Anamnestic
- (4) (2) & (3) both

161. If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidence :

- (1) Haemocytes
- (2) Serum albumines
- (3) Serum globulins
- (4) Fibrinogen in plasma

162. Small proteins produced by vertebrate cells naturally in response to viral infections and which inhibit the multiplication of viruses are called :

- (1) Immunoglobulins
- (2) Interferons
- (3) Antitoxins
- (4) Lipoproteins

163. Monocytes move from the systemic circulatory system into general connective tissues, where they differentiate into what phagocytic cell type?

- (1) Macrophage
- (2) T cell
- (3) B cell
- (4) Neutrophil

164. Complete the following analogy T-cell : cell mediated :: B-cell

- (1) Primary
- (2) Secondary
- (3) Humoral
- (4) Inflammatory

165. How does vaccination work :

- (1) The immune system produces antibodies which not stay in blood
- (2) Memory lymphocyte are produced. They remain in the blood to fight off any future infection with the live pathogen
- (3) The tissue macrophage stay in the blood vessel and constantly stimulate immune system
- (4) All of these

166. In which one of the following options, the two examples are correctly matched with their particular type of immunity :

Example	Type of immunity
(1) Polymorpho nuclear leucocytes and monocytes	Cellular barrier
(2) Anti-tetanus and anti-snake bite injection	Active immunity
(3) Saliva in mouth and tears in eyes	Physical barreirs
(4) Mucus coating of epithelium of urino-genital tract and HCl in stomach	Physiological barrier

167. Short lived immunity acquired from mother to foetus across placenta or through mother's milk to the infant is characterized as :

- (1) Active immunity
- (2) Passive immunity
- (3) Cellular immunity
- (4) Innate non-specific immunity

168. Which of the following organs is primarily concerned with immunity :
- (1) Liver
 - (2) Lymphatic tissue
 - (3) Kidney
 - (4) Thyroid
169. The cells which suppress the entire immune system from its attack in the same body are known as :
- (1) Helper T-cells
 - (2) killer B-cells
 - (3) Suppressor B-cells
 - (4) Suppressor T-cells
170. Vaccination protects a person from disease because it :
- (1) Helps in better digestion
 - (2) Increases RBC count
 - (3) Produces antibodies
 - (4) Corrects body heating system
171. B-cells are stimulated to produce antibodies by -
- (1) Killer cells
 - (2) Suppressor cells
 - (3) Cytotoxic cells
 - (4) Helper cells
172. Interferons are -
- (1) Anti bacterial proteins
 - (2) Anti viral proteins
 - (3) Anti helminthic proteins
 - (4) Anti fungal proteins
173. In which lymphoid organ, immature lymphocytes differentiate into antigen sensitive lymphocytes?
- (1) Spleen and tonsils
 - (2) Peyer's patches and MALT
 - (3) Thymus and bone marrow
 - (4) Appendix and spleen
174. The immunity provided to new born infant by colostrum is -
- (1) Active, Natural
 - (2) Active, Artificial
 - (3) Passive, Artificial
 - (4) Passive , Natural
175. Cell mediated immunity is provided by -
- (1) B- lymphocytes
 - (2) T- Lymphocytes
 - (3) Plasma cells
 - (4) B- Memory cells
176. When killed or attenuated pathogens are introduced during vaccination, immunological response is generated along with memory cells. It is an example of
- (1) Artificial acquired active immunity
 - (2) Artificial acquired passive immunity
 - (3) Natural acquired active immunity
 - (4) Natural acquired passive immunity
177. Which of the following are examples of passive immunity?
- (i) Presence of IgA in colostrum
 - (ii) Anti-tetanus serum
 - (iii) Tetanus toxoid vaccination
 - (iv) Antibiotic injection to patient with pneumonia
 - (v) Anti - venom given to patients with snake bite
- (1) (i), (ii) and (v)
 - (2) (i), (ii), (iii) and (v)
 - (3) (ii), (iii) and (iv)
 - (4) (ii), (iii) and (v)
178. Mark the correct statement regarding innate immunity.
- (1) Interferons constitute the cytokine barrier of immunity
 - (2) Tears and saliva constitute the anatomical barrier
 - (3) B- and T-lymphocytes form the cellular barrier
 - (4) Interferons protect against specific bacterial infections
179. Which of the following cells can non - specifically kill virus infected cells and some tumour cells of the body by creating perforin lined pores in the plasma membrane of the target cells?
- (1) Helper T-lymphocytes
 - (2) Cytotoxic T-cells
 - (3) B - lymphocytes
 - (4) Natural killer cells
180. Factors like body temperature, low pH in stomach, tears in eyes and various body secretions, prevent growth of many pathogens in body. These factors are included in
- (1) Natural passive immunity
 - (2) Artificial passive immunity
 - (3) Innate physiological barrier
 - (4) Innate anatomical barrier

Space For Rough Work

Result : NEET (UG) 2019

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AIR

✓ 8 in Top 10 All India Ranks

✓ 61 in Top 100 All India Ranks

TOTAL STUDENTS QUALIFIED

Classroom
63415

95102*

Distance
31687

1

NALIN KHANDELWAL
Classroom

AIR 2  BHAVIK BANSAL Distance	AIR 4  SWASTIK BHATIA Distance	AIR 5  ANANT JAIN Classroom	AIR 6  SARTHAK R. BHAT Distance	AIR 8  DHRUV KUSHWAHA Distance	AIR 9  MIHIR RAI Distance	AIR 10  RAGHAV DUBEY Classroom									
AIR-11  HEMANT KHANDELWAL Distance	AIR-17  RIDHAM AGARWAL Distance	AIR-18  HARSHVI N. J. Classroom	AIR-19  ARUNANGSHU B. Classroom	AIR-20  SATVIK LODHA Classroom	AIR-21  HARSH JAIN Distance	AIR-22  HARDIK GUPTA Classroom	AIR-24  LALIT AGARWAL Classroom	AIR-26  APOORV BAGHAV Classroom	AIR-27  PARTH GILL Distance	AIR-28  GAUTAM CHELLANI Distance	AIR-30  HARSH AGARWAL Classroom	AIR-37  LOHITASHYA GOYAL Distance	AIR-38  VISHWA RAKESH Distance	AIR-39  ANURAG GAUTAM Distance	AIR-44  M. H. SAYAN SHAH Classroom
AIR-45  SHIKHAR GUPTA Distance	AIR-46  DIPRASOM DAS Distance	AIR-47  BAHADUR SINGH Distance	AIR-48  AMAN AGRAWAL Classroom	AIR-49  VARUN CHOPRA Distance	AIR-51  ISHAAN MISHRA Distance	AIR-52  DISHA S. AGRAWAL Distance	AIR-53  SARVAGYA JAIN Distance	AIR-54  HARRY GOYAL Distance	AIR-56  ISHIKA GUPTA Distance	AIR-58  SHREYAS AGRAWAL Classroom	AIR-60  ARNAY MITTAL Classroom	AIR-61  CHIRAG JORVEKAR Distance	AIR-63  VINEET HARSH Distance	AIR-65  VAMSHI K. NEELA Distance	AIR-66  SHREY MUDGAL Classroom
AIR-69  PRIYANSHU AGARWAL Classroom	AIR-70  ISHIKA KAMRA Classroom	AIR-71  STUTI KHANDWALA Classroom	AIR-72  NAVEEN SHARMA Distance	AIR-75  SARTHAK SHARMA Classroom	AIR-76  VANSHIKA ARORA Distance	AIR-79  SAYAM V. BHATT Classroom	AIR-81  UDITH MOHAN Distance	AIR-84  PRAJNAPAN BASU Distance	AIR-85  HARSH RAMOLIYA Distance	AIR-86  UTKARSH SHARMA Classroom	AIR-87  NISHTHA Distance	AIR-89  ARYAN BANSAL Distance	AIR-91  SANGEETA AGRAWAL Classroom	AIR-92  YASHDEEP Classroom	AIR-93  ISHAN GUPTA Classroom
AIR-94  MEGHA GUPTA Classroom	AIR-96  GAUTAM KAUSHAL Classroom	AIR-97  POOJA YADAV Distance											AIR-99  PRAGYA MITRA Distance	AIR-100  VIVEK KUMAR Classroom	

*As Per Result Compiled so far

Authenticity of Result :
Power of **ALLEN**



Heartiest
Congratulations!